

GROUP-GRADED ORE EXTENSIONS AND GRADED SIMPLICITY

YASSINE AIT MOHAMED

ABSTRACT. We study Ore extensions $R[t; \sigma, \delta]$ of rings graded by an arbitrary group Γ . The grading of R extends to a grading with t homogeneous of degree γ if and only if σ and δ are compatible with conjugation and translation by γ , and this grading is then unique. For nonabelian Γ the degree of t need not centralize the support of R ; this twist can be removed by a change of variable when R_γ contains a unit, but not in general. The graded forms of the classical necessary conditions for simplicity are not sufficient, as the quantized Weyl algebra shows. If R is graded-simple and σ is an automorphism, then $R[t; \sigma, \delta]$ is graded-simple if and only if it contains no homogeneous normal element that is monic of positive t -degree. For Ore extensions of group algebras, an affine action of the group on the line describes all graded ideals; in characteristic zero, graded simplicity is equivalent to the nonvanishing of a class in $H^1(G, k_\chi)$, and simplicity forces the group to be metabelian.

1. INTRODUCTION

Ore extensions are the noncommutative polynomial rings in one variable [26]. They include skew polynomial rings, rings of differential operators and quantized Weyl algebras. Iterated Ore extensions provide many of the standard examples of noncommutative Noetherian rings [10, 21]. Two questions about an Ore extension $R[t; \sigma, \delta]$ recur: which structures on R extend to it, and when is it simple? The second has a long history, from Amitsur's work on differential polynomial rings over simple rings [2] and Jordan's thesis [13] to Cauchon, Lam-Leroy and Öinert-Richter-Silvestrov [4, 17, 25].

This paper studies both questions for gradings. Gradings of Ore extensions are usually taken over $\mathbb{Z}$: σ preserves degrees and δ raises them by one [28]. Many natural rings, however, are graded by nonabelian groups. A group algebra kG is graded by G , and so is every crossed product of a ring by G . For nonabelian gradings a new phenomenon appears. The defining relation $tr = \sigma(r)t + \delta(r)$ compares tr with $\sigma(r)t$, not with rt . Hence σ can absorb conjugation by the degree of t , and gradings appear that have no analogue over abelian groups.

The differential case $\sigma = \text{id}$ was treated in [1]. There the degree of t must centralize the support of R , and in characteristic zero gr-simplicity is governed by the δ -stable graded ideals of R and by the outeriness of δ . Both statements fail for general Ore extensions; we find the correct replacements.

All rings are associative with identity and nonzero, and ring homomorphisms preserve the identity. Ideals are two-sided unless stated otherwise. The letter k always denotes a field. We write $\text{char } R = 0$ if $n1_R \neq 0$ for every integer $n \geq 1$.

Let R be a ring and σ a ring endomorphism of R . A σ -derivation is an additive map $\delta: R \rightarrow R$ such that $\delta(rs) = \delta(r)s + \sigma(r)\delta(s)$ for all $r, s \in R$. The Ore extension $A = R[t; \sigma, \delta]$ is the free left R -module with basis $1, t, t^2, \dots$, with multiplication determined by

$$tr = \sigma(r)t + \delta(r) \quad (r \in R). \quad (1)$$

2020 Mathematics Subject Classification. 16S36, 16W50, 16W25, 16D30, 16S35, 17B63, 20J06.

Key words and phrases. Ore extension, skew derivation, group-graded ring, gr-simple ring, group algebra, group cohomology, skew group ring, Poisson polynomial ring, semiclassical limit.

We write $R[t; \sigma]$ when $\delta = 0$ and $R[t; \delta]$ when $\sigma = \text{id}$. For the general theory see [10, 21].

Every $f \in A$ can be written uniquely as $f = \sum_i a_i t^i$ with $a_i \in R$. If $f \neq 0$, its t -degree $\deg_t f$ is the largest n with $a_n \neq 0$, and $\text{lc}(f) = a_n$ is its *leading coefficient*. The t -degree should not be confused with the Γ -degree of a homogeneous element. For $n \geq 1$, the element f is *monic of t -degree n* if $f = t^n + \sum_{i < n} a_i t^i$. An element f of a ring S is *normal* if $Sf = fS$; in that case Sf is an ideal.

For $u \in R$, the map

$$\delta_u(r) = ur - \sigma(r)u \quad (r \in R)$$

is a σ -derivation. A σ -derivation is *inner* if it equals δ_u for some $u \in R$, and *outer* otherwise. If σ is an automorphism, each $\sigma^j \delta \sigma^{-j}$ is again a σ -derivation (see the proof of Lemma 2.3(3)). If σ is an automorphism and $n \geq 1$, we put

$$\Delta_n = \sum_{j=0}^{n-1} \sigma^j \delta \sigma^{-j},$$

a σ -derivation. Note that $\Delta_n = n\delta$ when $\sigma = \text{id}$.

Throughout, Γ is a group with identity e , not necessarily abelian. A Γ -graded ring is a ring $R = \bigoplus_{\tau \in \Gamma} R_\tau$ with $R_\rho R_\tau \subseteq R_{\rho\tau}$ for all ρ, τ ; see [22]. The elements of R_τ are *homogeneous of degree τ* , and $\text{Supp}(R) = \{\tau \in \Gamma : R_\tau \neq 0\}$ is the *support* of R . An ideal is *graded* if it is generated by homogeneous elements. A graded ring is *gr-simple* if 0 and the ring itself are its only graded ideals.

For $\gamma \in \Gamma$ put $\iota_\gamma(\tau) = \gamma\tau\gamma^{-1}$. An additive map $f: R \rightarrow R$ is ι_γ -graded if $f(R_\tau) \subseteq R_{\iota_\gamma(\tau)}$ for all $\tau \in \Gamma$. If γ is central, this is the usual notion of a graded map (compare [12, Section 1.1]).

Definition 1.1. Let R be Γ -graded and $\gamma \in \Gamma$. A pair (σ, δ) , where σ is a ring endomorphism of R and δ a σ -derivation, is γ -compatible if

$$\sigma(R_\tau) \subseteq R_{\gamma\tau\gamma^{-1}} \quad \text{and} \quad \delta(R_\tau) \subseteq R_{\gamma\tau} \quad (\tau \in \Gamma). \quad (2)$$

In that case δ is γ -inner if $\delta = \delta_u$ for some $u \in R_\gamma$, and γ -outer otherwise.

Thus the first condition in (2) says that σ is ι_γ -graded, and the second that δ is homogeneous of degree γ , acting on degrees by left translation. We show in Lemma 2.3(2) that a γ -compatible δ is γ -inner if and only if it is inner.

An ideal I of R is (σ, δ) -stable if $\sigma(I) \subseteq I$ and $\delta(I) \subseteq I$. A graded ring R is (σ, δ) -gr-simple if 0 and R are its only (σ, δ) -stable graded ideals.

Let G be a group. A *character* is a group homomorphism $\chi: G \rightarrow k^\times$. Let k_χ denote k as a G -module via $g \cdot \lambda = \chi(g)\lambda$; see [3] for group cohomology.

Definition 1.2. A *crossed homomorphism* (or *1-cocycle*) for χ is a map $c: G \rightarrow k$ with

$$c(\tau\rho) = c(\tau) + \chi(\tau)c(\rho) \quad (\tau, \rho \in G).$$

The crossed homomorphisms form a k -vector space $Z^1(G, k_\chi)$. The *coboundaries* are the cocycles $x(\chi - 1)$ with $x \in k$; they form a subspace $B^1(G, k_\chi)$, and

$$H^1(G, k_\chi) = Z^1(G, k_\chi) / B^1(G, k_\chi).$$

For $\chi = 1$ this is $H^1(G, k)$ with trivial coefficients, and $Z^1(G, k_1) = \text{Hom}(G, k)$.

The *affine group of the line* is the group $\text{Aff}(1, k) \cong k \rtimes k^\times$ of maps $X \mapsto aX + b$ with $a \in k^\times$ and $b \in k$, under composition. A group is *metabelian* if it has an abelian normal subgroup with abelian quotient; every subgroup of $\text{Aff}(1, k)$ is metabelian. If G acts on

a ring S by automorphisms θ_τ , the skew group ring $S \rtimes_\theta G = \bigoplus_{\tau \in G} Su_\tau$ has multiplication $(fu_\tau)(gu_\rho) = f\theta_\tau(g)u_{\tau\rho}$, and it is G -graded by $\deg u_\tau = \tau$.

A Poisson algebra is a commutative k -algebra P with a Lie bracket $\{-, -\}$ that is a derivation in each variable. It is Γ -graded if $P_\rho P_\tau \subseteq P_{\rho\tau}$ and $\{P_\rho, P_\tau\} \subseteq P_{\rho\tau}$ for all ρ, τ . A Poisson ideal is an ideal I with $\{P, I\} \subseteq I$, and P is Poisson gr-simple if 0 and P are its only graded Poisson ideals. An element $f \in P$ is Poisson normal if $\{P, f\} \subseteq fP$.

A Poisson derivation of P is a derivation α with $\alpha(\{a, b\}) = \{\alpha(a), b\} + \{a, \alpha(b)\}$. Given α , a Poisson α -derivation is a derivation δ with

$$\delta(\{a, b\}) = \{\delta(a), b\} + \{a, \delta(b)\} + \alpha(a)\delta(b) - \delta(a)\alpha(b) \quad (a, b \in P).$$

For such a pair, Oh [23] constructed the Poisson polynomial ring $B = P[t; \alpha, \delta]_P$: the unique Poisson bracket on $P[t]$ that extends the bracket of P and satisfies $\{t, a\} = \alpha(a)t + \delta(a)$ (see Section 4). For $u \in P$ put $\delta_u^P(a) = \{u, a\} - u\alpha(a)$. We say that δ is α -inner if $\delta = \delta_u^P$ for some $u \in P$. If Γ is abelian and $\gamma \in \Gamma$, the pair (α, δ) is γ -compatible if $\alpha(P_\tau) \subseteq P_\tau$ and $\delta(P_\tau) \subseteq P_{\gamma\tau}$ for all τ . Finally, P is (α, δ) -Poisson gr-simple if 0 and P are its only graded Poisson ideals I with $\alpha(I) + \delta(I) \subseteq I$.

Let S be a ring and $h \in S$ a central non-zero-divisor such that S/hS is commutative. Then S/hS is a Poisson algebra with bracket $\{\bar{a}, \bar{b}\} = \overline{h^{-1}(ab - ba)}$, called the semiclassical limit of S .

Fix a Γ -graded ring R and $\gamma \in \Gamma$. When does the grading of R extend to a Γ -grading of $R[t; \sigma, \delta]$ with $\deg t = \gamma$? In the differential case, tr and rt must have the same degree, so γ commutes with every element of $\text{Supp}(R)$ [1, Propositions 2.1 and 2.2]. In general σ may absorb conjugation by γ .

Theorem A (Theorem 2.1). *Let $\gamma \in \Gamma$. The Ore extension $A = R[t; \sigma, \delta]$ admits a Γ -grading extending the grading of R with $t \in A_\gamma$ if and only if (σ, δ) is γ -compatible. Such a grading is unique, and*

$$A_\alpha = \bigoplus_{n \geq 0} R_{\alpha\gamma^{-n}} t^n \quad (\alpha \in \Gamma).$$

For abelian Γ , the conditions (2) say that σ preserves degrees and δ shifts them by γ . For $\Gamma = \mathbb{Z}$ and $\gamma = 1$ they define the usual graded Ore extensions, and for $\sigma = \text{id}$ they recover the criterion of [1]. For nonabelian Γ new gradings appear: σ may move homogeneous components, and γ need not centralize $\text{Supp}(R)$ (Example 2.6). This twist can be removed when R_γ contains a unit (Remark 2.2(3)), but not in general (Example 3.13). To our knowledge, this extension problem has not been considered before.

Next we consider gr-simplicity. If A is gr-simple, then R is (σ, δ) -gr-simple and δ is outer (Proposition 3.1). These are the graded forms of the standard necessary conditions for simplicity [25, Proposition 4.6 and Remark 4.1]. For $\sigma = \text{id}$ they are also sufficient in characteristic zero. This is due to Jordan for the trivial grading [13], and it holds for arbitrary gradings by [1, Theorem 1.10(2)]. Once $\sigma \neq \text{id}$, sufficiency fails already for the trivial grading: over a commutative domain, a simple Ore extension has $\sigma = \text{id}$ [25, Theorem 4.4]. The failure persists for gr-simplicity. For the quantized Weyl algebra $\mathbb{Q}\langle x, t \rangle / (tx - 2xt - 1)$, graded by $\deg x = 1$ and $\deg t = -1$ and viewed as the Ore extension $\mathbb{Q}[x][t; \sigma, \delta]$, the ring $\mathbb{Q}[x]$ is (σ, δ) -gr-simple and δ is outer, but A is not gr-simple (Proposition 3.2).

The obstruction is the homogeneous normal element $1 + xt = tx - xt$, the classical normal element of the quantized Weyl algebra [14]. It exists because the commutator of t with $a_n t^n + \dots$ contains the term $(\sigma(a_n) - a_n)t^{n+1}$. So commuting with t no longer lowers the t -degree, and the leading-coefficient argument of the differential case breaks down (Remark

3.3). Our next result shows that, over a gr-simple coefficient ring, homogeneous normal elements are the only obstruction. Normal and semi-invariant polynomials in Ore extensions have been studied [6, 7, 15, 18, 20], and the maps Δ_n already appear in the study of normal elements [8, Lemma 4.8].

Theorem B (Theorem 3.7, Corollaries 3.8 and 3.10). *Let (σ, δ) be γ -compatible, and assume that R is gr-simple and that σ is an automorphism.*

- (1) *A is gr-simple if and only if A contains no homogeneous normal element that is monic of positive t -degree.*
- (2) *If Δ_n is outer for every $n \geq 1$, then A is gr-simple.*
- (3) *If $\text{char } R = 0$ and $\sigma\delta\sigma^{-1} - \delta$ is inner, then A is gr-simple if and only if δ is outer.*

Part (1) holds for every grading group and in every characteristic. Part (3) applies in particular when $\sigma\delta = \delta\sigma$. For the trivial grading, the ideals of Ore extensions over quasi-simple rings go back to Cauchon [4], and part (3) is close to a theorem of Chuang and Tsai on semi-invariant polynomials over prime rings [6] (Remark 3.11). The graded statement differs because a gr-simple ring need not be prime: the group algebra kC_2 , with $C_2 = \langle g \rangle$, is gr-simple for its natural grading, but $(1+g)(1-g) = 0$.

Group algebras give explicit examples, with a complete description of the graded ideals. Let G be a group, $\gamma \in G$ and $\chi: G \rightarrow k^\times$ a character. Grade kG by G and put $\sigma(\tau) = \chi(\tau)\gamma\tau\gamma^{-1}$. The k -linear σ -derivations compatible with $\deg t = \gamma$ are exactly the maps $\delta_c(\tau) = c(\tau)\gamma\tau$ with $c \in Z^1(G, k_\chi)$ (Proposition 2.7). The cocycle identity for c says precisely that

$$\Phi: G \rightarrow \text{Aff}(1, k), \quad \varphi_\tau(X) = \chi(\tau)X + c(\tau),$$

is a group homomorphism. This affine action controls the whole ideal theory.

Theorem C (Propositions 2.7 and 3.18, Theorem 3.19). *Let $A = kG[t; \sigma, \delta_c]$ with $\deg t = \gamma$, and let $\bar{k}$ be an algebraic closure of k .*

- (1) *δ_c is inner if and only if $c \in B^1(G, k_\chi)$.*
- (2) *Put $s = \gamma^{-1}t$. Then A is isomorphic, as a G -graded ring, to the skew group ring $k[X] \rtimes G$ for the action of G by the affine substitutions φ_τ^{-1} .*
- (3) *A is gr-simple if and only if $\Phi(G)$ has no finite orbit in $\bar{k}$, if and only if the restriction of $[c]$ to every subgroup of finite index is nonzero. If $\text{char } k = 0$, this holds if and only if $[c] \neq 0$ in $H^1(G, k_\chi)$.*
- (4) *A is simple if and only if it is gr-simple and Φ is injective. In particular, if A is simple, then G is metabelian.*

The graded ideals of A are 0 and the ideals $Af(s)$, where f runs over the monic polynomials with $f \circ \varphi_\tau \in k^\times f$ for all τ (Theorem 3.19(1)). In positive characteristic the cohomological criterion fails. No extension of this type over a finite group is gr-simple, although $H^1(G, k_\chi)$ can be nonzero, already for $G = C_p$ (Corollary 3.21). For $G = S_3 \cong \text{Aff}(1, \mathbb{F}_3)$ the obstruction is carried by an orbit rather than by a fixed point (Example 3.20 and Remark 3.22). Part (4) rests on a theorem of Öinert on skew group rings [24].

Parts (3) and (4) separate gr-simplicity from simplicity. For the free group $F(a, b)$ and $\gamma = a$ one obtains the gr-simple ring

$$ta = qat, \quad tb = aba^{-1}t + ab \quad (q \in \mathbb{Q} \setminus \{0, 1\}),$$

which is not simple (Examples 3.17 and 3.23). The subgroup of $\text{Aff}(1, \mathbb{Q})$ generated by $X \mapsto 2X$ and $X \mapsto X + 1$ gives a simple one.

In these examples t has the noncentral degree a , but the change of variable $s = \gamma^{-1}t$ moves it to e (Remark 2.2(3)). A degree that cannot be moved requires R_γ to contain no unit, as in Example 2.6, which is not gr-simple, and in Example 3.13, which is: there Γ is the fundamental group of the Klein bottle, σ is not graded, and A contains the first Weyl algebra.

The same questions make sense for Poisson algebras. Let Γ be abelian and P a Γ -graded Poisson algebra, and let $B = P[t; \alpha, \delta]_P$. Since P is commutative, nonzero products and brackets involve only commuting degrees, and the twist ι_γ of Theorem A disappears (see the beginning of Section 4).

Theorem D (Proposition 4.1, Theorem 4.3). *Let $\gamma \in \Gamma$.*

- (1) *B admits a grading extending that of P with $t \in B_\gamma$ if and only if (α, δ) is γ -compatible.*
- (2) *Assume that this holds and that P is Poisson gr-simple. Then B is Poisson gr-simple if and only if B contains no homogeneous Poisson normal element $t^n + \sum_{i < n} b_i t^i$ with $n \geq 1$.*
- (3) *If moreover $\text{char } k = 0$, then B is Poisson gr-simple if and only if δ is not α -inner.*

Part (3) needs no analogue of the innerness hypothesis of Theorem B(3). The reason is that the bracket $\{a, t^n\}$ produces $n\delta(a)$ directly, whereas in the skew case the corresponding coefficient is Δ_n . Semiclassical limits connect the two settings. The quantized Weyl algebra of Proposition 3.2 lies in the graded family $zx = qxz + (q - 1)$, whose semiclassical limit is the Poisson polynomial ring with $\{z, x\} = xz + 1$. The element $1 + xz$ is normal in the family and Poisson normal in the limit (Example 4.6). Moreover, if $\delta = \delta_u$ is inner, then its semiclassical limit is α -inner (Lemma 4.5).

The paper is organized as follows. In Section 2 we prove Theorem A, study γ -compatible pairs and construct the nonabelian examples. Section 3 is devoted to gr-simplicity; it proves Proposition 3.2 and Theorems B and C. Finally, Section 4 treats Poisson polynomial rings and semiclassical limits, and proves Theorem D.

2. GRADINGS ON ORE EXTENSIONS

Throughout this section, σ is a ring endomorphism of R , δ is a σ -derivation, and $A = R[t; \sigma, \delta]$.

Theorem 2.1. *Let $\gamma \in \Gamma$. The following are equivalent:*

- (i) *A admits a Γ -grading $\{A_\alpha\}$ with $R_\tau \subseteq A_\tau$ for all τ and $t \in A_\gamma$.*
- (ii) *(σ, δ) is γ -compatible.*

In that case the grading in (i) is unique, namely

$$A_\alpha = \bigoplus_{n \geq 0} R_{\alpha\gamma^{-n}} t^n \quad (\alpha \in \Gamma). \quad (3)$$

Moreover, if $\delta \neq 0$, then γ is determined by δ .

Proof. (ii) $\Rightarrow$ (i). Define A_α by (3). Since $A = \bigoplus_n R t^n$ is free over R and $R = \bigoplus_\rho R_\rho$, we have $A = \bigoplus_{n, \rho} R_\rho t^n$. Placing $R_\rho t^n$ in $A_{\rho\gamma^n}$ gives $A = \bigoplus_\alpha A_\alpha$, and $1 \in A_e$. It remains to prove $A_\alpha A_\beta \subseteq A_{\alpha\beta}$.

First, $R_\rho A_\beta \subseteq A_{\rho\beta}$, because $R_\rho R_{\beta\gamma^{-n}} \subseteq R_{\rho\beta\gamma^{-n}}$. Next, $tA_\beta \subseteq A_{\gamma\beta}$: for $r \in R_\rho$ and $n \geq 0$ we have $rt^n \in A_{\rho\gamma^n}$, and by (1) and (2),

$$t r t^n = \sigma(r) t^{n+1} + \delta(r) t^n \in R_{\rho\gamma^{-1}} t^{n+1} + R_{\rho\gamma^n} t^n \subseteq A_{\gamma\rho\gamma^n}.$$

By induction on m , $t^m A_\beta \subseteq A_{\gamma^m \beta}$ for all $m \geq 0$: if it holds for m , then $t^{m+1} A_\beta \subseteq t A_{\gamma^m \beta} \subseteq A_{\gamma^{m+1} \beta}$. Hence $(rt^m) A_\beta \subseteq R_\rho A_{\gamma^m \beta} \subseteq A_{\rho\gamma^m \beta}$ for $r \in R_\rho$. Since A_α is spanned by such rt^m with $\rho\gamma^m = \alpha$, the grading is multiplicative.

(i) $\Rightarrow$ (ii). Let $0 \neq r \in R_\tau$, and write $\sigma(r) = \sum_\rho s_\rho$ and $\delta(r) = \sum_\rho d_\rho$ with $s_\rho, d_\rho \in R_\rho$. By (i), $s_\rho t \in A_{\rho\gamma}$, $d_\rho \in A_\rho$ and $tr \in A_{\gamma\tau}$. For $\beta \neq \gamma\tau$, the component of $tr = \sum_\rho s_\rho t + \sum_\rho d_\rho$ of degree β is $s_{\beta\gamma^{-1}}t + d_\beta$, and it vanishes. Since $1, t$ are R -linearly independent, $s_{\beta\gamma^{-1}} = 0 = d_\beta$. Hence $\sigma(r) \in R_{\gamma\tau\gamma^{-1}}$ and $\delta(r) \in R_{\gamma\tau}$.

Uniqueness. Let $\{A'_\alpha\}$ be a grading as in (i). Then $t^n \in A'_{\gamma^n}$, so $R_\rho t^n \subseteq A'_{\rho\gamma^n}$ and $A_\alpha \subseteq A'_\alpha$ for every α . If $a \in A'_\alpha$, write $a = \sum_\beta a_\beta$ with $a_\beta \in A_\beta \subseteq A'_\beta$; uniqueness of homogeneous components in $\{A'_\beta\}$ gives $a = a_\alpha \in A_\alpha$. Finally, if $\delta \neq 0$, choose a homogeneous $x \in R_\tau$ with $\delta(x) \neq 0$. If γ and γ' both satisfy (2), then $0 \neq \delta(x) \in R_{\gamma\tau} \cap R_{\gamma'\tau}$, so $\gamma = \gamma'$. $\square$

Remark 2.2. (1) For nonabelian Γ , σ is ι_γ -graded but need not be graded: $\sigma(R_\tau) \subseteq R_\tau$ for all τ if and only if $\gamma\tau\gamma^{-1} = \tau$ whenever $\sigma(R_\tau) \neq 0$. In Examples 2.6 and 3.17, σ is not graded.

(2) If $\sigma = \text{id}$, then $R_\tau \subseteq R_{\gamma\tau\gamma^{-1}}$ forces $\gamma\tau\gamma^{-1} = \tau$ for $\tau \in \text{Supp}(R)$. So γ centralizes $\text{Supp}(R)$ and $\delta(R_\tau) \subseteq R_{\gamma\tau} = R_{\tau\gamma}$. This recovers [1, Proposition 2.1].

(3) Let (σ, δ) be γ -compatible and let $g \in R_\gamma$ be a unit. Then $g^{-1} \in R_{\gamma^{-1}}$ (compare components of $gg^{-1} = 1$). Put $t' = g^{-1}t \in A_e$, $\sigma'(r) = g^{-1}\sigma(r)g$ and $\delta' = g^{-1}\delta$. Then $t'r = \sigma'(r)t' + \delta'(r)$, σ' is a ring endomorphism, and

$$\delta'(rr') = g^{-1}(\sigma(r)\delta(r') + \delta(r)r') = \sigma'(r)\delta'(r') + \delta'(r)r',$$

so δ' is a σ' -derivation. Moreover $\sigma'(R_\tau) \subseteq R_{\gamma^{-1}}R_{\gamma\tau\gamma^{-1}}R_\gamma \subseteq R_\tau$ and $\delta'(R_\tau) \subseteq R_{\gamma^{-1}}R_{\gamma\tau} \subseteq R_\tau$. By induction, $t'^n = u_n t^n + (\text{lower terms})$, where

$$u_n = g^{-1}\sigma(g^{-1}) \cdots \sigma^{n-1}(g^{-1})$$

is a unit; so $\{t'^n\}_{n \geq 0}$ is a left R -basis of A , and $A = R[t'; \sigma', \delta']$ with $\deg t' = e$ and σ' graded. In particular, if $\sigma(r) = grg^{-1}$ for all r , then $\sigma' = \text{id}$ and A is the graded differential polynomial ring $R[t'; \delta']$. Thus a degree of t that cannot be moved to e in this way requires R_γ to contain no unit (Examples 2.6 and 3.13).

Lemma 2.3. Let (σ, δ) be γ -compatible.

- (1) For $u \in R_\gamma$, the pair (σ, δ_u) is γ -compatible.
- (2) δ is γ -inner if and only if $\delta = \delta_u$ for some $u \in R$.
- (3) If σ is an automorphism, then $\sigma^j(R_\tau) = R_{\gamma^j\tau\gamma^{-j}}$ for all $j \in \mathbb{Z}$ and $\tau \in \Gamma$, the pair $(\sigma, \sigma^j\delta\sigma^{-j})$ is γ -compatible for every $j \in \mathbb{Z}$, and $\sigma\delta_u\sigma^{-1} = \delta_{\sigma(u)}$ with $\sigma(u) \in R_\gamma$ for $u \in R_\gamma$.

Proof. (1) For $r \in R_\tau$ we have $ur \in R_{\gamma\tau}$ and $\sigma(r)u \in R_{\gamma\tau\gamma^{-1}}R_\gamma \subseteq R_{\gamma\tau}$, so $\delta_u(r) \in R_{\gamma\tau}$.

(2) Suppose $\delta = \delta_u$ with $u = \sum_\rho u_\rho$, and let $r \in R_\tau$. Then $u_\rho r \in R_{\rho\tau}$ and $\sigma(r)u_\rho \in R_{\gamma\tau\gamma^{-1}\rho}$. Each of these degrees equals $\gamma\tau$ only for $\rho = \gamma$. Since $\delta(r) \in R_{\gamma\tau}$, comparing components of degree $\gamma\tau$ in $\delta(r) = \sum_\rho (u_\rho r - \sigma(r)u_\rho)$ gives $\delta(r) = u_\gamma r - \sigma(r)u_\gamma$. By additivity, $\delta = \delta_{u_\gamma}$.

(3) We first show $\sigma(R_\tau) = R_{\gamma\tau\gamma^{-1}}$. The inclusion $\subseteq$ is compatibility. Let $x \in R_{\gamma\tau\gamma^{-1}}$ and write $x = \sigma(y)$ with $y = \sum_\rho y_\rho$. The terms $\sigma(y_\rho) \in R_{\rho\gamma\gamma^{-1}}$ lie in pairwise distinct degrees, so $x = \sigma(y_\tau)$. Hence $\sigma(R_\tau) = R_{\gamma\tau\gamma^{-1}}$ and, σ being bijective, $\sigma^{-1}(R_\rho) = R_{\gamma^{-1}\rho\gamma}$. Induction on $|j|$ gives $\sigma^j(R_\tau) = R_{\gamma^j\tau\gamma^{-j}}$ for all $j \in \mathbb{Z}$.

Put $\delta_j = \sigma^j\delta\sigma^{-j}$. Since $\sigma^{\pm j}$ are ring homomorphisms,

$$\delta_j(ab) = \sigma^j(\sigma^{1-j}(a)\delta(\sigma^{-j}(b)) + \delta(\sigma^{-j}(a))\sigma^{-j}(b)) = \sigma(a)\delta_j(b) + \delta_j(a)b,$$

so δ_j is a σ -derivation. For $a \in R_\tau$ we get $\sigma^{-j}(a) \in R_{\gamma^{-j}\tau\gamma^j}$, then $\delta\sigma^{-j}(a) \in R_{\gamma^{1-j}\tau\gamma^j}$, and finally $\delta_j(a) \in R_{\gamma^j\gamma^{1-j}\tau\gamma^j\gamma^{-j}} = R_{\gamma\tau}$. Finally, applying σ to $\delta_u(\sigma^{-1}(r)) = u\sigma^{-1}(r) - ru$ gives $\sigma\delta_u\sigma^{-1}(r) = \sigma(u)r - \sigma(r)\sigma(u) = \delta_{\sigma(u)}(r)$, and $\sigma(u) \in R_{\gamma\gamma\gamma^{-1}} = R_\gamma$. $\square$

By Lemma 2.3(2), a γ -compatible σ -derivation is γ -inner if and only if it is inner. From now on we simply say *inner* or *outer*. As in the ungraded case [9, Lemma 1.5], the inner case reduces to a skew polynomial ring.

Lemma 2.4. *Let (σ, δ) be γ -compatible with $\delta = \delta_u$, $u \in R_\gamma$. Then $s = t - u \in A_\gamma$ satisfies $sr = \sigma(r)s$ for $r \in R$, $A = \bigoplus_{n \geq 0} Rs^n$ is the graded skew polynomial ring $R[s; \sigma]$ with $\deg s = \gamma$, and As is a nonzero proper graded ideal of A .*

Proof. Since $\delta(r) = ur - \sigma(r)u$,

$$sr = \sigma(r)t + \delta(r) - ur = \sigma(r)t - \sigma(r)u = \sigma(r)s.$$

By (1), $t^i u \in \sigma^i(u)t^i + \sum_{m < i} Rt^m$, and induction on n gives $s^n - t^n \in \sum_{m < n} Rt^m$. Hence $\{s^n\}_{n \geq 0}$ is a left R -basis of A . Since $s(rs^n) = \sigma(r)s^{n+1} \in As$, we get $sA \subseteq As$, so the left ideal As is two-sided. In the basis $\{s^n\}$ it equals $\bigoplus_{n \geq 1} Rs^n$, so $1 \notin As$. It is graded because s is homogeneous. $\square$

Example 2.5 (Quantized Weyl algebra). Let $R = \mathbb{Q}[x]$ with $\deg x = 1$, $\sigma(f)(x) = f(2x)$ and $\delta(f) = (f(2x) - f(x))/x$. Then $\delta(fg) = \sigma(f)\delta(g) + \delta(f)g$, σ preserves degrees, and

$$\delta(x^m) = (2^m - 1)x^{m-1} \quad (m \geq 1).$$

Thus (σ, δ) is (-1) -compatible, and by Theorem 2.1

$$A = \mathbb{Q}[x][t; \sigma, \delta] \cong \mathbb{Q}\langle x, t \rangle / (tx - 2xt - 1), \quad \deg x = 1, \deg t = -1.$$

In the next example σ moves homogeneous components and t has noncentral degree.

Example 2.6 (A free algebra graded by S_3). Compose permutations from right to left. Let $\Gamma = S_3$, $\gamma = (123)$, and let $R = k\langle x_1, x_2, x_3 \rangle$ be the free algebra graded by

$$\deg x_1 = (12), \quad \deg x_2 = (23), \quad \deg x_3 = (13).$$

Then $\gamma(12)\gamma^{-1} = (23)$, $\gamma(23)\gamma^{-1} = (13)$, $\gamma(13)\gamma^{-1} = (12)$, and $\gamma(12) = (13)$, $\gamma(23) = (12)$, $\gamma(13) = (23)$. Let σ be the automorphism $x_1 \mapsto x_2 \mapsto x_3 \mapsto x_1$, and let δ be the σ -derivation with

$$\delta(x_1) = x_3, \quad \delta(x_2) = x_1, \quad \delta(x_3) = x_2;$$

it exists and is unique because R is free. Thus $\deg \sigma(x_i) = \gamma \deg(x_i) \gamma^{-1}$ and $\deg \delta(x_i) = \gamma \deg(x_i)$. Both properties pass to products: if $u \in R_\alpha$ and $v \in R_\beta$ satisfy them, then $\sigma(uv) \in R_{\gamma\alpha\beta\gamma^{-1}}$ and $\delta(uv) = \sigma(u)\delta(v) + \delta(u)v \in R_{\gamma\alpha\beta}$. Hence (σ, δ) is γ -compatible, and $A = R[t; \sigma, \delta]$ is graded with $\deg t = (123)$.

The transpositions generate S_3 , whose centre is trivial, so no $\gamma \neq e$ centralizes $\text{Supp}(R)$. By Remark 2.2(2), a differential polynomial ring over R admits a compatible grading only with $\deg t = e$. The homogeneous units of R are the nonzero scalars, so R_γ contains no unit and Remark 2.2(3) does not apply.

The derivation δ is outer. A monomial of length ℓ has degree of sign $(-1)^\ell$, and γ is even, so every monomial in R_γ has even length $\ell \geq 2$. For $u \in R_\gamma$, every monomial of $\delta_u(x_1) = ux_1 - x_2u$ therefore has length at least 3, whereas $\delta(x_1) = x_3$. Hence $\delta \neq \delta_u$, and Lemma 2.3(2) applies. On the other hand, the ideal generated by x_1, x_2, x_3 is graded and stable under σ and δ , so A is not gr-simple by Proposition 3.1 below.

The degree of t cannot be moved to e : no $s \in A_e$ has powers forming a left R -basis of A . Suppose that $s \in A_e$ is such an element. Since R is a domain and σ is injective, $\deg_t(fg) = \deg_t f + \deg_t g$ for nonzero $f, g \in A$. Then $d = \deg_t s \geq 1$ because $A \neq R$, and every nonzero element of $\bigoplus_n Rs^n$ has t -degree divisible by d ; applied to t , this gives $d = 1$. Write $s = bt + a$ and $t = r_1s + r_0$ with $a, b, r_0, r_1 \in R$. By (3), $b \in R_{\gamma^{-1}}$, and comparing

t -coefficients gives $r_1 b = 1$. In the free algebra this forces b to be a nonzero scalar, so $b \in R_e$. This contradicts $b \in R_{\gamma^{-1}}$ with $\gamma^{-1} \neq e$.

We now turn to group algebras, in the notation of Definition 1.2. Taking $\tau = \rho = e$ in the cocycle identity gives $c(e) = 0$, and then $c(\tau^{-1}) = -\chi(\tau)^{-1}c(\tau)$.

Proposition 2.7. *Let k be a field, G a group, $R = kG$ with $R_\tau = k\tau$, $\gamma \in G$, and $\chi: G \rightarrow k^\times$ a homomorphism. Put $\sigma(\tau) = \chi(\tau)\gamma\tau\gamma^{-1}$.*

- (1) σ extends to a k -algebra automorphism of kG with $\sigma(R_\tau) = R_{\gamma\tau\gamma^{-1}}$.
- (2) The k -linear σ -derivations δ for which (σ, δ) is γ -compatible are exactly the maps $\delta_c(\tau) = c(\tau)\gamma\tau$ with $c \in Z^1(G, k_\chi)$.
- (3) δ_c is γ -inner if and only if $c \in B^1(G, k_\chi)$.
- (4) $\sigma\delta_c\sigma^{-1} - \delta_c = -\delta_{c(\gamma)\gamma}$. In particular it is γ -inner.

Proof. (1) For $\tau, \rho \in G$,

$$\sigma(\tau)\sigma(\rho) = \chi(\tau)\chi(\rho)\gamma\tau\gamma^{-1}\gamma\rho\gamma^{-1} = \sigma(\tau\rho).$$

So $\tau \mapsto \sigma(\tau)$ is a group homomorphism into the units of kG , and it extends k -linearly to an algebra endomorphism of kG . Since $k^\times$ is abelian, $\chi(\gamma^{-1}\tau\gamma) = \chi(\tau)$, and $\psi(\tau) = \chi(\tau)^{-1}\gamma^{-1}\tau\gamma$ satisfies $\sigma\psi(\tau) = \tau = \psi\sigma(\tau)$. Hence $\psi = \sigma^{-1}$. Finally $\sigma(k\tau) = k\chi(\tau)\gamma\tau\gamma^{-1} = k\gamma\tau\gamma^{-1}$ because $\chi(\tau) \neq 0$.

(2) Compatibility forces $\delta(\tau) \in R_{\gamma\tau} = k\gamma\tau$, say $\delta(\tau) = c(\tau)\gamma\tau$. Then

$$\sigma(\tau)\delta(\rho) + \delta(\tau)\rho = \chi(\tau)\gamma\tau\gamma^{-1}c(\rho)\gamma\rho + c(\tau)\gamma\tau\rho = (c(\tau) + \chi(\tau)c(\rho))\gamma\tau\rho.$$

Since G is a k -basis of kG , the identity $\delta(\tau\rho) = \sigma(\tau)\delta(\rho) + \delta(\tau)\rho$ is exactly the cocycle identity.

Conversely, for a cocycle c the k -linear map δ_c satisfies the Leibniz rule on the basis G , hence everywhere by bilinearity.

(3) Write $u = x\gamma \in R_\gamma$ with $x \in k$. Then

$$\delta_u(\tau) = x\gamma\tau - \chi(\tau)\gamma\tau\gamma^{-1}x\gamma = x(1 - \chi(\tau))\gamma\tau.$$

So the γ -inner δ_c are those with $c = -x(\chi - 1)$, that is, $c \in B^1(G, k_\chi)$.

(4) We have $\sigma^{-1}(\tau) = \chi(\tau)^{-1}\gamma^{-1}\tau\gamma$ and $\sigma(\tau\gamma) = \chi(\tau)\chi(\gamma)\gamma\tau$. Hence

$$\sigma\delta_c\sigma^{-1}(\tau) = \chi(\tau)^{-1}c(\gamma^{-1}\tau\gamma)\sigma(\tau\gamma) = \chi(\gamma)c(\gamma^{-1}\tau\gamma)\gamma\tau.$$

The cocycle identity, applied to $\gamma^{-1} \cdot (\tau\gamma)$ and to $\tau \cdot \gamma$, together with $c(\gamma^{-1}) = -\chi(\gamma)^{-1}c(\gamma)$, gives

$$\chi(\gamma)c(\gamma^{-1}\tau\gamma) = -c(\gamma) + c(\tau) + \chi(\tau)c(\gamma) = c(\tau) - (1 - \chi(\tau))c(\gamma).$$

By the formula in (3) with $x = c(\gamma)$, this reads $\sigma\delta_c\sigma^{-1}(\tau) = \delta_c(\tau) - \delta_{c(\gamma)\gamma}(\tau)$. Since $c(\gamma)\gamma \in R_\gamma$, the difference is γ -inner. $\square$

Every k -linear ι_γ -graded endomorphism of kG is of the form $\tau \mapsto \chi(\tau)\gamma\tau\gamma^{-1}$ for a character χ : it maps the unit τ to a nonzero element of $k\gamma\tau\gamma^{-1}$, and it is multiplicative. Hence Proposition 2.7 describes all γ -compatible pairs of k -linear maps on kG . If $\chi = 1$, then σ is conjugation by the homogeneous unit γ , and Remark 2.2(3) identifies $kG[t; \sigma, \delta_c]$ with the differential polynomial ring $kG[t'; \delta']$, where $\deg t' = e$, $\delta'(\tau) = c(\tau)\tau$ and $c \in \text{Hom}(G, k)$. If $\chi \neq 1$, then σ is not conjugation by a homogeneous unit: the homogeneous units are the elements $\lambda\rho$ with $\lambda \in k^\times$, $\rho \in G$, and conjugation by $\lambda\rho$ maps τ to $\rho\tau\rho^{-1}$, whereas $\sigma(\tau) = \chi(\tau)\gamma\tau\gamma^{-1}$. Hopf–Ore extensions of group algebras [27] are related, but no Hopf structure is used here.

3. GR-SIMPLICITY

Throughout this section (σ, δ) is γ -compatible and A carries the grading (3).

3.1. Necessary conditions and a counterexample.

Proposition 3.1. *If A is gr-simple, then R is (σ, δ) -gr-simple and δ is outer.*

Proof. Let I be a graded ideal of R with $\sigma(I) + \delta(I) \subseteq I$, and put $IA = \bigoplus_n It^n$. Clearly $R(IA) \subseteq IA$ and $(IA)t \subseteq IA$. Iterating (1) gives $t^n r \in \sum_{m \leq n} Rt^m$, so $(it^n)r \in \sum_m It^m$ for $i \in I$ and $r \in R$. Finally $t(it^n) = \sigma(i)t^{n+1} + \delta(i)t^n \in IA$. Since R and t generate A , IA is an ideal of A . It is graded by (3), and $IA \cap R = I$. Hence $I \in \{0, R\}$ when A is gr-simple. If δ is inner, it is γ -inner by Lemma 2.3(2), and Lemma 2.4 gives a nonzero proper graded ideal. $\square$

For $\sigma = \text{id}$ the converse holds in characteristic zero [1, Theorem 1.10(2)]. For skew derivations it fails.

Proposition 3.2. *In Example 2.5, $R = \mathbb{Q}[x]$ is (σ, δ) -gr-simple and δ is outer, but A is not gr-simple.*

Proof. The nonzero proper graded ideals of $\mathbb{Q}[x]$ are the ideals (x^m) with $m \geq 1$, and none is δ -stable since $\delta(x^m) = (2^m - 1)x^{m-1}$. As $R_{-1} = 0$, the only (-1) -inner σ -derivation is 0, while $\delta(x) = 1$; so δ is outer by Lemma 2.3(2).

Let $z = 1 + xt \in A_0$. From $tx = 2xt + 1$ we get

$$zx = x + x(2xt + 1) = 2xz, \quad tz = t + (2xt + 1)t = 2zt.$$

Hence $zA = Az$, and Az is a nonzero graded ideal. The assignment $x \mapsto X$, $t \mapsto -X^{-1}$ kills $tx - 2xt - 1$, so it defines a homomorphism $A \rightarrow \mathbb{Q}[X^{\pm 1}]$. It kills z and is the identity on $\mathbb{Q}$, so $Az \neq A$. $\square$

Remark 3.3. The element z comes from a localization. On the gr-simple ring $R' = \mathbb{Q}[x^{\pm 1}]$ the maps σ and δ extend by the same formulas, and $\delta = \delta_u$ with $u = -x^{-1} \in R'_{-1}$, since both are σ -derivations and $\delta_u(x) = -1 + 2 = 1 = \delta(x)$. By Lemma 2.4, $s = t + x^{-1}$ generates a proper graded ideal of $R'[t; \sigma, \delta]$, and $z = xs$. Thus δ is outer on R only because the element that makes it inner is not in R . The element z is normal of t -degree one (see [15] for such elements and [14] for localizations of the quantized Weyl algebra).

The leading-coefficient argument of [1] fails because, for $f = a_n t^n + a_{n-1} t^{n-1} + \dots$,

$$tf - ft = (\sigma(a_n) - a_n)t^{n+1} + (\delta(a_n) + \sigma(a_{n-1}) - a_{n-1})t^n + \dots$$

The commutator with t no longer lowers the t -degree, and ideals of leading coefficients need not be δ -stable. Here the leading coefficients of the elements of Az of t -degree one form the ideal (x) : if $a_1 t + a_0 \in Az$, applying the homomorphism above gives $a_1 = xa_0$, and conversely $a_0 z = a_0 + xa_0 t$. The ideal (x) is not δ -stable.

3.2. A characterization over gr-simple coefficient rings. From now on σ is an automorphism. By Lemma 2.3(3), $\sigma(R_\tau) = R_{\iota_\gamma(\tau)}$ for every τ . The proof of Theorem B rests on ideals of leading coefficients; for $\sigma = \text{id}$ and the trivial grading, compare [25, Lemma 4.13]. For a graded ideal J of A and $n \geq 0$ let

$$L_n(J) = \left\{ a \in R : at^n + \sum_{i < n} a_i t^i \in J \text{ for some } a_i \in R \right\}.$$

Lemma 3.4. *$L_n(J)$ is a graded ideal of R .*

Proof. Let $a \in L_n(J)$, witnessed by $f = at^n + \sum_{i < n} a_i t^i \in J$, and let $r \in R$. Then $rf \in J$ has t^n -coefficient ra , so $ra \in L_n(J)$. By (1), $t^i \sigma^{-n}(r) \in \sigma^{i-n}(r)t^i + \sum_{m < i} Rt^m$ for every $i \geq 0$. Hence $f\sigma^{-n}(r) \in J$ has t -degree at most n and t^n -coefficient ar , so $ar \in L_n(J)$.

For gradedness, write $a = \sum_{\tau} a_{\tau}$. By (3), the component of f of degree $\tau\gamma^n$ lies in J , has t -degree at most n , and has t^n -coefficient a_{τ} . So $a_{\tau} \in L_n(J)$ for every τ . $\square$

Lemma 3.5. For $n \geq 1$ and $r \in R$,

$$t^n r = \sigma^n(r)t^n + D_n(r)t^{n-1} + \sum_{m < n-1} Rt^m, \quad D_n = \sum_{j=0}^{n-1} \sigma^j \delta \sigma^{n-1-j}.$$

Proof. For $n = 1$ this is (1). Assume the formula for n and multiply it on the left by t . By (1),

$$t^{n+1}r = \sigma^{n+1}(r)t^{n+1} + (\delta\sigma^n(r) + \sigma D_n(r))t^n + \sum_{m < n} Rt^m,$$

since $t(D_n(r)t^{n-1}) = \sigma D_n(r)t^n + \delta D_n(r)t^{n-1}$ and the remaining terms have t -degree at most $n-1$. Finally $\delta\sigma^n + \sigma D_n = \sum_{j=0}^n \sigma^j \delta \sigma^{n-j} = D_{n+1}$. $\square$

The next lemma records basic properties of homogeneous normal elements that are monic of positive t -degree.

Lemma 3.6. Let $n \geq 1$ and let $f = t^n + at^{n-1} + \sum_{i < n-1} a_i t^i$ be homogeneous. Then $f \in A_{\gamma^n}$ and $a \in R_{\gamma}$. Moreover:

- (1) f is normal if and only if $fr = \sigma^n(r)f$ for all $r \in R$ and $tf - ft \in Rf$. In that case $tf = ft + (\sigma(a) - a)f$.
- (2) If f is normal, then $\Delta_n = \delta_{-a}$. In particular Δ_n is inner.
- (3) If f is normal, then Af is a nonzero proper graded ideal of A .

Proof. The t^n -coefficient 1 lies in R_e , so $f \in A_{\gamma^n}$ by (3), and then $a \in R_{\gamma^n\gamma^{1-n}} = R_{\gamma}$. Since f is monic, $\text{lc}(gf) = \text{lc}(g)$ for every nonzero $g \in A$, hence

$$\deg_t(gf) = \deg_t(g) + n \quad (g \neq 0). \quad (4)$$

(1) Suppose f is normal and let $r \in R$. Write $fr = gf$ with $g \in A$. By (4), $\deg_t g \leq 0$, so $g \in R$, and comparing t^n -coefficients via Lemma 3.5 gives $g = \sigma^n(r)$. Similarly $ft = hf$ with $\deg_t h = 1$; comparing t^{n+1} -coefficients gives $h = t + h_0$ with $h_0 \in R$, so $tf - ft = -h_0f \in Rf$. Conversely, assume both conditions and write $tf - ft = cf$ with $c \in R$. Then $fr = \sigma^n(r)f$ and $ft = (t - c)f$, so $fA \subseteq Af$ because R and t generate A . Since σ^n is bijective, $rf = f\sigma^{-n}(r)$ and $tf = ft + f\sigma^{-n}(c)$, so $Af \subseteq fA$. Finally $tf = t^{n+1} + \sigma(a)t^n + \dots$ and $ft = t^{n+1} + at^n + \dots$, and comparing t^n -coefficients in $tf - ft = cf$ gives $c = \sigma(a) - a$.

(2) By (1), $fr = \sigma^n(r)f$. Compare t^{n-1} -coefficients. By Lemma 3.5, $t^n r$ contributes $D_n(r)$ and $at^{n-1}r$ contributes $a\sigma^{n-1}(r)$; the terms $a_i t^i r$ with $i < n-1$ do not contribute. On the right, $\sigma^n(r)f$ contributes $\sigma^n(r)a$. Hence

$$D_n(r) + a\sigma^{n-1}(r) = \sigma^n(r)a.$$

Substitute $r = \sigma^{1-n}(y)$. Since $D_n\sigma^{1-n} = \sum_{j=0}^{n-1} \sigma^j \delta \sigma^{-j} = \Delta_n$, this becomes

$$\Delta_n(y) = \sigma(y)a - ay = (-a)y - \sigma(y)(-a) = \delta_{-a}(y).$$

(3) $Af = fA$ is an ideal, graded because f is homogeneous, and nonzero. By (4) its nonzero elements have t -degree at least $n \geq 1$, so $1 \notin Af$. $\square$

Theorem 3.7. Assume that R is gr-simple and σ is an automorphism. Then A is gr-simple if and only if A contains no homogeneous normal element that is monic of positive t -degree.

Proof. If such an element exists, A is not gr-simple by Lemma 3.6(3). Conversely, let $J \neq 0$ be a graded ideal of A , and let n be the least t -degree of a nonzero element of J . By Lemma 3.4, $L_n(J)$ is a nonzero graded ideal of R , so $L_n(J) = R$. Choose $g \in J$ of t -degree n with t^n -coefficient 1. If $n = 0$, then $g = 1 \in J$ and $J = A$. Let $n \geq 1$. The component of g of degree γ^n lies in J and has the form

$$f = t^n + at^{n-1} + \sum_{i < n-1} a_i t^i.$$

For $r \in R$, both fr and $\sigma^n(r)f$ have t^n -coefficient $\sigma^n(r)$. So $fr - \sigma^n(r)f \in J$ has t -degree less than n , and it vanishes by minimality of n . Likewise, in $tf - ft - (\sigma(a) - a)f \in J$ the t^{n+1} -terms cancel and the t^n -coefficients agree, so this element vanishes. By Lemma 3.6(1), f is a homogeneous normal element, monic of positive t -degree. Hence, if no such element exists, $J = A$. $\square$

For the trivial grading, Lam and Leroy characterize the simplicity of general Ore extensions [17, Theorems 4.5 and 5.8], and for $\sigma = \text{id}$ the argument of Theorem 3.7 is that of [25, Proposition 4.14]. See also [4, 7] for ideals of Ore extensions and [18, 20, 6] for semi-invariant polynomials. The corollaries below give conditions on δ that exclude the obstruction of Theorem 3.7; for the trivial grading they are simplicity criteria over simple coefficient rings.

Corollary 3.8. *Assume that R is gr-simple and σ is an automorphism. If Δ_n is outer for every $n \geq 1$, then A is gr-simple.*

Proof. Combine Theorem 3.7 and Lemma 3.6(2). $\square$

The argument of Lemma 3.6(2) is the one in [8, proof of Lemma 4.8], where the coefficient ring is a connected graded algebra generated in degree one, without zero divisors in degree one, whose only normal elements are the scalars.

Corollary 3.8 never applies to $\sigma = \text{id}$ in characteristic $p > 0$, since then $\Delta_p = p\delta = 0$. Theorem 3.7 has no such restriction, and for $\sigma = \text{id}$ its argument works under a weaker hypothesis on R .

Corollary 3.9. *Let $\sigma = \text{id}$, let δ be compatible with $\deg t = \gamma$, and assume that R is δ -gr-simple. Then $R[t; \delta]$ is gr-simple if and only if it contains no homogeneous central element that is monic of positive t -degree.*

Proof. A central element is normal, so one direction is Lemma 3.6(3). Conversely, let $J \neq 0$ be a graded ideal and let n be the least t -degree of a nonzero element of J . By Lemma 3.4, $L_n(J)$ is a nonzero graded ideal of R . If $f = \sum_i a_i t^i \in J$ with $\deg_t f \leq n$, then $tf - ft = \sum_i \delta(a_i) t^i \in J$, so $\delta(L_n(J)) \subseteq L_n(J)$. Hence $L_n(J) = R$. If $n = 0$, then $1 \in J$. Let $n \geq 1$. As in the proof of Theorem 3.7, J contains a homogeneous element $f = t^n + \sum_{i < n} a_i t^i$. For $r \in R$, the element $fr - rf \in J$ has t -degree less than n , and so does $tf - ft = \sum_{i < n} \delta(a_i) t^i$, because $\delta(1) = 0$. Both vanish by minimality of n , so f is central. $\square$

Corollary 3.9 is the graded form of [25, Lemma 4.13 and Proposition 4.14]. It needs no hypothesis on Γ or on the homogeneous elements of A , in contrast with the centre criterion [1, Theorem 1.10(3)]. Compare also [25, Theorem 4.15]: $R[t; \delta]$ is simple if and only if R is δ -simple and the centre of $R[t; \delta]$ is a field.

Corollary 3.10. *Assume that R is gr-simple with $\text{char } R = 0$, that σ is an automorphism, and that $\sigma\delta\sigma^{-1} - \delta$ is inner. Then A is gr-simple if and only if δ is outer. This applies in particular when $\sigma\delta = \delta\sigma$.*

Proof. Necessity is Proposition 3.1. For sufficiency, write $\sigma\delta\sigma^{-1} - \delta = \delta_v$. This map is compatible by Lemma 2.3(3), so we may take $v \in R_\gamma$ by Lemma 2.3(2).

We show by induction that $\sigma^j\delta\sigma^{-j} - \delta = \delta_{v_j}$ with $v_j \in R_\gamma$ for all $j \geq 0$. For $j = 0$ take $v_0 = 0$. If it holds for j , then, since $u \mapsto \delta_u$ is additive and $\sigma\delta_{v_j}\sigma^{-1} = \delta_{\sigma(v_j)}$ with $\sigma(v_j) \in R_\gamma$ by Lemma 2.3(3),

$$\sigma^{j+1}\delta\sigma^{-j-1} - \delta = \sigma(\sigma^j\delta\sigma^{-j} - \delta)\sigma^{-1} + (\sigma\delta\sigma^{-1} - \delta) = \delta_{\sigma(v_j)+v}.$$

Summing over $j = 0, \dots, n-1$ gives $\Delta_n = n\delta + \delta_{w_n}$ with $w_n = \sum_{j < n} v_j \in R_\gamma$.

Let $n \geq 1$. The element $n1_R$ is central, homogeneous of degree e , and nonzero since $\text{char } R = 0$. So $(n1_R)R$ is a nonzero graded ideal, hence equal to R , and $n1_R$ is invertible with central inverse n^{-1} . If $\Delta_n = \delta_y$ for some $y \in R$, then $n\delta = \delta_{y-w_n}$, and multiplying by n^{-1} gives $\delta = \delta_{n^{-1}(y-w_n)}$. This contradicts the outerness of δ . Hence every Δ_n is outer, and Corollary 3.8 applies. $\square$

Remark 3.11. Let R be prime, σ an automorphism and δ an X -outer σ -derivation, that is, δ is not δ_b with b in the symmetric Martindale ring of quotients of R [6]. Recall that $f \in R[t; \sigma, \delta]$ is *semi-invariant* if $fR \subseteq Rf$. Chuang and Tsai [6] show that, in characteristic zero, a semi-invariant polynomial of minimal positive degree, when one exists, has degree equal to the least integer $\nu \geq 1$ for which

$$\sigma^\nu\delta\sigma^{-\nu} - \delta$$

is X -inner. No monic semi-invariant polynomial of degree one exists: if $(t+b)r = r'(t+b)$ for each r , then comparing t -coefficients gives $r' = \sigma(r)$, and comparing constant terms gives $\delta(r) = \sigma(r)b - br$, so δ would be X -inner. Thus, when $\nu = 1$, no nonconstant semi-invariant polynomial exists. For the trivial grading and R simple, Corollary 3.10 is therefore close to their result. It does not require R prime: kC_2 is gr-simple for its natural grading, but $(1+g)(1-g) = 0$.

Remark 3.12. The innerness hypothesis in Corollary 3.10 cannot simply be removed from the argument. Suppose that $\sigma^2 = \text{id}$, $\sigma\delta\sigma^{-1} = -\delta$, and $\delta^2(r) = rb - br$ for all $r \in R$, where $b \in R_{\gamma^2}$ satisfies $\sigma(b) = b$ and $\delta(b) = 0$. Then $D_2 = \delta\sigma + \sigma\delta = 0$ in Lemma 3.5, so $t^2r = rt^2 + \delta^2(r)$. The homogeneous element $f = t^2 + b$ satisfies $fr = rf$ and $tf - ft = (\sigma(b) - b)t + \delta(b) = 0$, so A is not gr-simple by Lemma 3.6. Yet, if R is gr-simple of characteristic zero, then $2 \cdot 1_R$ is a central unit (see the proof of Corollary 3.10), so $\sigma\delta\sigma^{-1} - \delta = -2\delta$ is outer whenever δ is. Such an example needs σ not to be conjugation by a unit of R_γ ; otherwise Remark 2.2(3) gives a graded differential polynomial ring, where outerness suffices in characteristic zero [1, Theorem 1.10(2)].

We now show that the twist of Theorem A can be intrinsic in a gr-simple extension.

Example 3.13 (A gr-simple extension with an intrinsic twist). Let $\Gamma = \langle h, r \mid rhr^{-1} = h^{-1} \rangle$; every element of Γ is uniquely $h^m r^n$ with $m, n \in \mathbb{Z}$. Let $R = \mathbb{Q}[h^{\pm 1}][y; \theta]$ with $\theta(h) = h^{-1}$, graded by $\deg h = h$ and $\deg y = r$. Thus $\{h^m y^i\}$ is a $\mathbb{Q}$ -basis of R , $h^m y^i \in R_{h^m r^i}$, and

$$(h^m y^i)(h^n y^j) = h^{m+(-1)^i n} y^{i+j}.$$

Put $\gamma = r^{-1}$ and define $\mathbb{Q}$ -linear maps by

$$\sigma(h^m y^i) = h^{-m} y^i, \quad \delta(h^m y^i) = i h^{-m} y^{i-1}.$$

Both sides of $\sigma(ab) = \sigma(a)\sigma(b)$ equal $h^{-m-(-1)^i n} y^{i+j}$ for $a = h^m y^i$, $b = h^n y^j$, so σ is an automorphism with $\sigma^2 = \text{id}$. Since $(-1)^{i-1} = -(-1)^i$,

$$\sigma(a)\delta(b) + \delta(a)b = j h^{-m-(-1)^i n} y^{i+j-1} + i h^{-m+(-1)^{i-1} n} y^{i+j-1} = \delta(ab),$$

so δ is a σ -derivation. As $\gamma h^m r^i \gamma^{-1} = h^{-m} r^i$ and $\gamma h^m r^i = h^{-m} r^{i-1}$, the pair (σ, δ) is γ -compatible, and $A = R[t; \sigma, \delta]$ is graded with $\deg t = r^{-1}$. Its defining relations are

$$yh = h^{-1}y, \quad th = h^{-1}t, \quad ty = yt + 1.$$

Here γ does not centralize $h \in \text{Supp}(R)$, so σ is not graded, and $R_\gamma = 0$.

The ring A is gr-simple. Let W be the subalgebra generated by y and t . The elements $h^m y^i t^j$ form a $\mathbb{Q}$ -basis of A , so $W \cong A_1(\mathbb{Q})$, and $h^m y^i t^j \in A_{h^m r^{i-j}}$ gives $A_{h^m r^n} = h^m W_n$ with $W_n = \text{span}\{y^i t^j : i - j = n\}$. Let $J \neq 0$ be a graded ideal and $0 \neq x \in J_{h^m r^n}$. Then $h^{-m}x \in J \cap W$, so $J \cap W$ is a nonzero ideal of the simple ring $A_1(\mathbb{Q})$. Hence $1 \in J$.

The degree of t cannot be moved to e . By (3), $A_e = \bigoplus_n \mathbb{Q} y^n t^n$. Since R is a domain and σ is injective, the argument of Example 2.6 applies: an element $s \in A_e$ whose powers form a left R -basis has t -degree one, so $s = b y t + a$ with $a, b \in \mathbb{Q}$, and $t = r_1 s + r_0$ forces $r_1 b y = 1$ in R , which is impossible.

Note that R is not gr-simple, since $Ry = yR$ is a proper graded ideal, so Theorem 3.7 does not apply.

3.3. Group algebras. The ring kG is gr-simple, because every nonzero homogeneous element is a unit. By Proposition 2.7(4), Corollary 3.10 therefore applies to the family of Proposition 2.7.

Corollary 3.14. *Let $\text{char } k = 0$, and let γ, χ, σ and $c \in Z^1(G, k_\chi)$ be as in Proposition 2.7. Then $kG[t; \sigma, \delta_c]$, graded by $\deg t = \gamma$, is gr-simple if and only if $[c] \neq 0$ in $H^1(G, k_\chi)$.*

Proof. Combine Corollary 3.10 with Proposition 2.7(3),(4). $\square$

The cohomological criterion can be tested on generators.

Corollary 3.15. *In the setting of Corollary 3.14 (so $\text{char } k = 0$), suppose that G is generated by $s_1, \dots, s_m$. Put $u_i = c(s_i)$ and $v_i = \chi(s_i) - 1$. If $\chi = 1$, then $kG[t; \sigma, \delta_c]$ is gr-simple if and only if some u_i is nonzero. If $\chi \neq 1$, it is gr-simple if and only if*

$$u_i v_j - u_j v_i \neq 0 \quad \text{for some } i, j.$$

Proof. By the cocycle identity and $c(s^{-1}) = -\chi(s)^{-1}c(s)$, two cocycles that agree on the generators agree on G . So $[c] = 0$ if and only if $u_i = x v_i$ for all i and some $x \in k$. If $\chi = 1$, all v_i vanish, and this means $u_i = 0$ for all i . If $\chi \neq 1$, some v_i is nonzero, and (u_i) is a multiple of (v_i) if and only if every minor $u_i v_j - u_j v_i$ vanishes. Now apply Corollary 3.14. $\square$

For a presentation $G = \langle s_1, \dots, s_m \mid w_1, \dots, w_r \rangle$, the admissible vectors (u_i) are the solutions of linear equations. Extend the prescribed values to a cocycle on the free group by $c(s_i^{-1}) = -\chi(s_i)^{-1}u_i$ and $c(g_1 g_2) = c(g_1) + \chi(g_1)c(g_2)$. This cocycle descends to G if and only if $c(w_j) = 0$ for every relator. Indeed, $\chi(w_j) = 1$, so $c(g w_j g^{-1}) = \chi(g)c(w_j)$ for every word g ; hence c vanishes on the normal closure N of the relators, and $c(gn) = c(g)$ for $n \in N$. Corollary 3.15 then tests gr-simplicity on the solution space.

Remark 3.16. If $\chi(z) \neq 1$ for some z in the centre of G , then $H^1(G, k_\chi) = 0$. Indeed, applying the cocycle identity to $z\tau = \tau z$ gives $c(z) + \chi(z)c(\tau) = c(\tau) + \chi(\tau)c(z)$, that is, $c(\tau) = x(\chi(\tau) - 1)$ with $x = c(z)(\chi(z) - 1)^{-1}$. So every cocycle is a coboundary. In particular,

if G is abelian and $\chi \neq 1$, then δ_c is inner and no ring in the family is gr-simple (Proposition 3.1). Gr-simple examples with $\chi \neq 1$ require a nonabelian group on whose centre χ is trivial.

Example 3.17. Let $G = F(a, b)$ be the free group on a, b , let $k = \mathbb{Q}$, fix $q \in \mathbb{Q}^\times$ with $q \neq 1$, and take $\gamma = a$, $\chi(a) = q$, $\chi(b) = 1$. A cocycle on a free group is determined by arbitrary values on the free generators; let c be the cocycle with $c(a) = 0$ and $c(b) = 1$. From $tr = \sigma(r)t + \delta_c(r)$ we get

$$ta = \chi(a)aaa^{-1}t + c(a)aa = qat, \quad tb = \chi(b)aba^{-1}t + c(b)ab = aba^{-1}t + ab.$$

So the ring is generated by $\mathbb{Q}F(a, b)$ and t subject to these relations, with $\deg t = a$. Every coboundary $x(\chi - 1)$ vanishes at b , so $[c] \neq 0$, and the ring is gr-simple by Corollary 3.14. More precisely, cocycles correspond to pairs $(c(a), c(b)) \in \mathbb{Q}^2$ and coboundaries to the pairs $(x(q - 1), 0)$; hence $H^1(F(a, b), \mathbb{Q}_\chi) \cong \mathbb{Q}$ via $[c] \mapsto c(b)$, and the ring attached to c is gr-simple exactly when $c(b) \neq 0$. Since $c(\gamma) = c(a) = 0$, Proposition 2.7(4) gives $\sigma\delta_c = \delta_c\sigma$, the commuting case of Corollary 3.10. Here $\sigma(R_b) = R_{aba^{-1}} \neq R_b$, and since a does not commute with b , no differential polynomial ring over $\mathbb{Q}F(a, b)$ admits a compatible grading with $\deg t = a$ (Remark 2.2(2)). The degree can nevertheless be moved to e by Remark 2.2(3) with $g = a$.

3.4. Group algebras as skew group rings. For group algebras every $R_\gamma = k\gamma$ contains the unit γ , so by Remark 2.2(3) the degree of the Ore variable can be moved to e , and the extension becomes a skew group ring over a polynomial ring. The noncentral degree of t in Example 3.17 reflects the choice of generator.

Keep the notation of Proposition 2.7 and put $A = kG[t; \sigma, \delta_c]$. For $\tau \in G$ let

$$\varphi_\tau(X) = \chi(\tau)X + c(\tau) \in k[X].$$

Since $\varphi_\tau(\varphi_\rho(X)) = \chi(\tau\rho)X + c(\tau) + \chi(\tau)c(\rho)$, the cocycle identity says exactly that $\varphi_{\tau\rho} = \varphi_\tau \circ \varphi_\rho$. Thus $\Phi: \tau \mapsto \varphi_\tau$ is a homomorphism from G to the affine group $\text{Aff}(1, k) = k \rtimes k^\times$, and every homomorphism $G \rightarrow \text{Aff}(1, k)$ arises in this way from a unique pair (χ, c) with $c \in Z^1(G, k_\chi)$. Moreover, $c = x(\chi - 1)$ if and only if $\varphi_\tau(-x) = -x$ for all τ ; hence $[c] = 0$ if and only if $\Phi(G)$ has a fixed point in k . Let G act on $k[X]$ by the k -algebra automorphisms $\theta_\tau(f) = f \circ \varphi_\tau^{-1}$, and let $k[X] \rtimes_\theta G = \bigoplus_\tau k[X]u_\tau$ be the corresponding skew group ring, so that $u_\tau f = \theta_\tau(f)u_\tau$ and $\deg u_\tau = \tau$.

Proposition 3.18. Put $s = \gamma^{-1}t \in A_e$. Then $s\tau = \tau\varphi_\tau(s)$ for $\tau \in G$, $A_e = k[s]$ is a polynomial ring in s , and $A_\tau = \tau k[s]$ for every $\tau \in G$. The map $f(X)u_\tau \mapsto f(s)\tau$ is an isomorphism of G -graded k -algebras $k[X] \rtimes_\theta G \rightarrow A$.

Proof. Since $\chi(\tau)$ and $c(\tau)$ are central scalars,

$$s\tau = \gamma^{-1}(\chi(\tau)\gamma\tau\gamma^{-1}t + c(\tau)\gamma\tau) = \tau(\chi(\tau)s + c(\tau)) = \tau\varphi_\tau(s).$$

By (1), $t\gamma^{-n} \in \chi(\gamma)^{-n}\gamma^{-n}t + R$, so by induction $s^n = \lambda_n\gamma^{-n}t^n + (\text{lower terms})$ with $\lambda_n \in k^\times$. Since $A_e = \bigoplus_n k\gamma^{-n}t^n$ by (3) and $s^n \in A_e$, the powers s^n form a k -basis of A_e . Since τ is a unit in A_τ , $\tau A_e \subseteq A_\tau$ and $\tau^{-1}A_\tau \subseteq A_e$, so $A_\tau = \tau k[s]$. From $\tau^{-1}s\tau = \varphi_\tau(s)$ we get $\tau f(s)\tau^{-1} = f(\varphi_\tau^{-1}(s)) = \theta_\tau(f)(s)$. Hence

$$f(s)\tau \cdot g(s)\rho = f(s)\theta_\tau(g)(s)\tau\rho,$$

which is the image of $(fu_\tau)(gu_\rho) = f\theta_\tau(g)u_{\tau\rho}$. So the map is multiplicative, and it maps $k[X]u_\tau$ bijectively onto $k[s]\tau = \tau k[s] = A_\tau$. $\square$

Skew group rings of group algebras over polynomial rings also appear in the study of iterated Hopf Ore extensions of group algebras [11].

Theorem 3.19. *Let $A = kG[t; \sigma, \delta_c]$ and $\Phi = (\chi, c)$ be as above, and let $\bar{k}$ be an algebraic closure of k .*

- (1) *The graded ideals of A are the ideals $Af(s)$, where $f = 0$ or $f \in k[X]$ is monic with $f \circ \varphi_\tau \in k^\times f$ for all $\tau \in G$. Distinct f give distinct ideals.*
- (2) *A is gr-simple if and only if $\Phi(G)$ has no finite orbit in $\bar{k}$, if and only if $[c]$ restricts to a nonzero class in $H^1(H, k_\chi)$ for every subgroup H of finite index in G .*
- (3) *If $\text{char } k = 0$, then A is gr-simple if and only if $[c] \neq 0$ in $H^1(G, k_\chi)$.*
- (4) *A is simple if and only if it is gr-simple and Φ is injective. In particular, if A is simple, then G is metabelian.*

Proof. (1) Let J be a graded ideal and $I = J \cap A_e$, an ideal of $A_e = k[s]$. For $\tau \in G$, $\tau I \subseteq J_\tau$ and $\tau^{-1}J_\tau \subseteq I$, so $J = \bigoplus_\tau \tau I$; moreover $\tau I \tau^{-1} \subseteq J \cap A_e = I$. Conversely, if I is an ideal of $k[s]$ with $\tau I \tau^{-1} \subseteq I$ for all τ , then $\bigoplus_\tau \tau I$ is an ideal of A , because $\rho g(s) \tau I = \rho \tau (\tau^{-1} g(s) \tau) I \subseteq \rho \tau I$ and $\tau I \rho g(s) = \tau \rho (\rho^{-1} I \rho) g(s) \subseteq \tau \rho I$. Thus $J \mapsto J \cap A_e$ is a bijection onto the ideals I of $k[s]$ with $\tau I \tau^{-1} \subseteq I$ for all τ . Write $I = f(s)k[s]$ with $f = 0$ or f monic. As $\tau f(s) \tau^{-1} = f(\varphi_\tau^{-1}(s))$ and $f \circ \varphi_\tau^{-1}$ has the same degree as f , the condition reads $f \circ \varphi_\tau^{-1} \in k^\times f$ for all τ , equivalently $f \circ \varphi_\tau \in k^\times f$ for all τ . Finally $\bigoplus_\tau \tau I = Af(s)$.

(2) By (1), A is gr-simple if and only if every monic f with $f \circ \varphi_\tau \in k^\times f$ for all τ equals 1. Let f be such a polynomial of positive degree, and let $Z \subseteq \bar{k}$ be its set of roots. If $z \in Z$, then $f \circ \varphi_\tau$ vanishes at $\varphi_\tau^{-1}(z)$, hence so does f . Thus Z is a finite nonempty $\Phi(G)$ -stable set, and its orbits are finite. Conversely, let $O \subseteq \bar{k}$ be a finite orbit, and let M be the finite set of minimal polynomials over k of the elements of O . For $x \in O$ and $\tau \in G$, $m_x \circ \varphi_\tau$ is irreducible of degree $\deg m_x$, because φ_τ is an affine bijection defined over k , and it vanishes at $\varphi_\tau^{-1}(x) \in O$; hence $m_x \circ \varphi_\tau \in k^\times m_{\varphi_\tau^{-1}(x)}$. The map $m_x \mapsto m_{\varphi_\tau^{-1}(x)}$ is well defined on M and surjective, since φ_τ^{-1} permutes O ; so it is a bijection of M . Hence $f = \prod_{m \in M} m$ is monic of positive degree with $f \circ \varphi_\tau \in k^\times f$ for all τ .

For the second equivalence, a finite orbit exists if and only if some subgroup H of finite index fixes a point $x \in \bar{k}$: take for H the stabilizer of a point of the orbit; conversely the orbit of x has at most $[G : H]$ elements. If H fixes x , then $c(h) = x(1 - \chi(h))$ for $h \in H$. If $\chi|_H \neq 1$, choosing $h_0 \in H$ with $\chi(h_0) \neq 1$ gives $x = c(h_0)/(1 - \chi(h_0)) \in k$, so $c|_H = (-x)(\chi - 1)$; if $\chi|_H = 1$, then $c|_H = 0$. In both cases the restriction of $[c]$ to H is zero. Conversely, if $c|_H = y(\chi - 1)$ with $y \in k$, then H fixes $-y$.

(3) Let $O \subseteq \bar{k}$ be a finite orbit, and put $b = |O|^{-1} \sum_{x \in O} x$, which is defined because $\text{char } k = 0$. Each φ_τ is affine and permutes O , so $\varphi_\tau(b) = b$. The argument of (2) with $H = G$ gives $[c] = 0$. Conversely, if $[c] = 0$, then $\Phi(G)$ fixes a point of k . Now apply (2).

(4) By Proposition 3.18, A is the skew group ring $k[X] \rtimes_\theta G$ over the commutative ring $k[X]$. By [24, Theorem 6.13], A is simple if and only if $k[X]$ is G -simple and maximal commutative in A . By the proof of (1), $k[X]$ is G -simple if and only if A is gr-simple. An element $\sum_\tau f_\tau u_\tau$ commutes with every $g \in k[X]$ if and only if $f_\tau(\theta_\tau(g) - g) = 0$ for all τ and g . As $k[X]$ is a domain, $k[X]$ is maximal commutative if and only if $\theta_\tau \neq \text{id}$ for $\tau \neq e$, that is, if and only if Φ is injective. Then $G \cong \Phi(G) \subseteq \text{Aff}(1, k) = k \rtimes k^\times$, which is metabelian. $\square$

Part (3) recovers Corollary 3.14 without using Corollary 3.10. The cohomological criterion fails in positive characteristic.

Example 3.20. Let $k = \mathbb{F}_3$, $G = \text{Aff}(1, \mathbb{F}_3) \cong S_3$, and let Φ be the inclusion; thus $\chi(\varphi) = a$ and $c(\varphi) = b$ for $\varphi(X) = aX + b$. No point of $\mathbb{F}_3$ is fixed by G , so $[c] \neq 0$ in $H^1(G, k_\chi)$. However, $f = X^3 - X$ satisfies $f(aX + b) = a^3 X^3 + b^3 - aX - b = af(X)$ for $a, b \in \mathbb{F}_3$. By Theorem 3.19(1), $A(s^3 - s)$ is a nonzero proper graded ideal of A for every choice of $\gamma \in G$. So Corollary 3.14 fails in characteristic 3.

More generally, the cohomological criterion fails for every finite group with $H^1(G, k_\chi) \neq 0$.

Corollary 3.21. *If G is finite, then $kG[t; \sigma, \delta_c]$ is not gr-simple. In particular, let $\text{char } k = p > 0$, $G = C_p = \langle g \rangle$, $\chi = 1$ and $c(g^i) = i \cdot 1_k$. Then $[c] \neq 0$ in $H^1(G, k)$, but $A(s^p - s)$, with $s = \gamma^{-1}t$ as in Proposition 3.18, is a nonzero proper graded ideal of A .*

Proof. Every $\Phi(G)$ -orbit in $\bar{k}$ has at most $|G|$ elements; apply Theorem 3.19(2). For $G = C_p$ and $\chi = 1$, the map c is a well-defined homomorphism because $p \cdot 1_k = 0$; moreover $B^1(G, k) = 0$ and $c \neq 0$, so $[c] \neq 0$. Since $c(\tau) \in \mathbb{F}_p$, the polynomial $f = X^p - X$ satisfies $f \circ \varphi_\tau = f$ for all τ , and Theorem 3.19(1) applies. $\square$

Here $\chi = 1$, so A is a differential polynomial ring by Remark 2.2(3). Thus the cohomological criterion already fails in the differential case.

Remark 3.22. Let $s = \gamma^{-1}t$, $\lambda \in k$, $G_\lambda = \{\tau \in G : \varphi_\tau(\lambda) = \lambda\}$ and $L_\lambda = A(s - \lambda)$. Then L_λ is a two-sided ideal if and only if λ is fixed by $\Phi(G)$. Indeed, by Remark 2.2(3) with $g = \gamma$, $A = kG[s; \sigma', \delta']$ with $\sigma'(\tau) = \chi(\tau)\tau$ and $\delta'(\tau) = c(\tau)\tau$, so

$$(s - \lambda)\tau = \sigma'(\tau)(s - \lambda) + (\varphi_\tau(\lambda) - \lambda)\tau \quad (\tau \in G).$$

Let D be the k -linear map with $D(\tau) = (\varphi_\tau(\lambda) - \lambda)\tau$. Since $s - \lambda$ is monic, $A = kG \oplus L_\lambda$. For $a = r + g(s - \lambda)$ with $r \in kG$, we have $L_\lambda a \subseteq L_\lambda$ if and only if $(s - \lambda)r \in L_\lambda$, that is, $D(r) \in kG \cap L_\lambda = 0$, that is, $r \in kG_\lambda$. Hence the idealizer $\{a \in A : L_\lambda a \subseteq L_\lambda\}$ equals $kG_\lambda \oplus L_\lambda$. Hence L_λ is two-sided if and only if its idealizer is A , that is, $G_\lambda = G$.

If $\text{char } k = 0$ and A is not gr-simple, then $[c] = 0$ by Theorem 3.19(3), so $\Phi(G)$ fixes some $\lambda \in k$ and L_λ is a nonzero proper graded ideal. In Example 3.20 this fails: every $\lambda \in \mathbb{F}_3$ has stabilizer $\{X \mapsto aX + \lambda(1 - a) : a = \pm 1\}$ of order 2, so no L_λ is two-sided, whereas $s^3 - s = \prod_{\lambda \in \mathbb{F}_3} (s - \lambda)$ generates a proper graded ideal. The obstruction is carried by the orbit $\mathbb{F}_3$ rather than by a point.

Example 3.23. (1) In Example 3.17, $\Phi(a)(X) = qX$ and $\Phi(b)(X) = X + 1$. The orbit of any $x \in \bar{k}$ contains $x + \mathbb{Z}$, so the ring is gr-simple, as already shown. Since $F(a, b)$ is not solvable and $\text{Aff}(1, \mathbb{Q})$ is metabelian, Φ is not injective, and the ring is not simple by Theorem 3.19(4).

(2) Let G be the subgroup of $\text{Aff}(1, \mathbb{Q})$ generated by $a(X) = 2X$ and $b(X) = X + 1$, let Φ be the inclusion, and let $\gamma = a$. Then $\chi(a) = 2$, $\chi(b) = 1$, $c(a) = 0$, $c(b) = 1$, and the orbit of every $x \in \bar{\mathbb{Q}}$ contains $x + \mathbb{Z}$. Here $\sigma(b) = aba^{-1} = b^2$. By Theorem 3.19(2),(4), the ring $QG[t; \sigma, \delta_c]$, with $ta = 2at$, $tb = b^2t + ab$ and $\deg t = a$, is simple.

4. GRADED POISSON POLYNOMIAL RINGS

Throughout this section k is a field and, except in Remark 4.7, Γ is abelian. In a commutative graded algebra, $P_\rho P_\tau \neq 0$ forces $\rho\tau = \tau\rho$, and since $\{P_\rho, P_\tau\} = \{P_\tau, P_\rho\} \subseteq P_{\rho\tau} \cap P_{\tau\rho}$, the same holds for nonzero brackets. In particular, if P is a domain, then $\text{Supp}(P)$ generates an abelian subgroup. Without this assumption the support need not commute: $k[x, y]/(xy)$ is graded by the free group on a, b with $\deg x = a$ and $\deg y = b$. In particular, the conjugation in (2) has no Poisson counterpart.

Let α be a Poisson derivation of P and δ a Poisson α -derivation. By Oh's theorem [23, Theorem 1.1], in the form of [19, Theorem 2.1], a Poisson bracket on the polynomial algebra $P[t]$ extends the bracket of P and satisfies

$$\{t, a\} = \alpha(a)t + \delta(a) \quad (a \in P)$$

if and only if α is a Poisson derivation and δ is a Poisson α -derivation; such a bracket is unique. We follow the sign convention of [19] and write $B = P[t; \alpha, \delta]_p$.

Proposition 4.1. *Let $\gamma \in \Gamma$. The Poisson algebra B admits a Γ -grading extending that of P with $t \in B_\gamma$ if and only if (α, δ) is γ -compatible. In that case the grading is unique, namely $B_\beta = \bigoplus_{n \geq 0} P_{\beta\gamma^{-n}} t^n$.*

Proof. If such a grading exists and $a \in P_\tau$, then $\{t, a\} = \alpha(a)t + \delta(a) \in B_{\gamma\tau}$. Comparing the coefficients of t and 1 in each homogeneous component, as in the proof of Theorem 2.1, gives $\alpha(a) \in P_\tau$ and $\delta(a) \in P_{\gamma\tau}$. Uniqueness is proved as in Theorem 2.1. Conversely, since Γ is abelian, the formula defines a grading of the commutative algebra $P[t]$. Call a pair (x, y) of homogeneous elements *good* if $\{x, y\} \in B_{\deg x \deg y}$. Pairs of homogeneous elements of P are good because P is graded, the pairs (t, a) and (a, t) are good by compatibility and antisymmetry, and (t, t) is good. If (x, y) and (x, z) are good, then so is (x, yz) , by the Leibniz rule and because Γ is abelian; the same holds in the first entry. Every homogeneous element of B is a sum of products of homogeneous elements of P and copies of t , all of the same degree, so every pair of homogeneous elements is good. $\square$

As in the associative case, innerness produces a Poisson normal element of t -degree one.

Lemma 4.2. *Let (α, δ) be γ -compatible.*

- (1) *If $\delta = \delta_u^P$ with $u = \sum_\rho u_\rho$, then $\delta = \delta_{u_\gamma}^P$.*
- (2) *If $\delta = \delta_u^P$ with $u \in P_\gamma$, then $t - u \in B_\gamma$ is Poisson normal and $B(t - u)$ is a nonzero proper graded Poisson ideal of B .*

Proof. (1) For $a \in P_\tau$, the elements $\{u_\rho, a\}$ and $u_\rho \alpha(a)$ lie in $P_{\rho\tau}$, while $\delta(a) \in P_{\gamma\tau}$. Comparing components of degree $\gamma\tau$ gives the claim.

(2) For $a \in P$, $\{t - u, a\} = \alpha(a)t + \delta(a) - \{u, a\} = \alpha(a)(t - u)$. Moreover $\delta(u) = \{u, u\} - u\alpha(u) = -u\alpha(u)$, so $\{t, t - u\} = -\{t, u\} = -\alpha(u)t - \delta(u) = -\alpha(u)(t - u)$. Since P and t generate B , the Leibniz rule gives $\{B, t - u\} \subseteq (t - u)B$. Hence $B(t - u)$ is a Poisson ideal. It is graded because $t - u$ is homogeneous, and proper because its nonzero elements have positive t -degree. $\square$

Theorem 4.3. *Let (α, δ) be γ -compatible and $B = P[t; \alpha, \delta]_P$, graded as in Proposition 4.1.*

- (1) *If B is Poisson gr-simple, then P is (α, δ) -Poisson gr-simple and δ is not α -inner.*
- (2) *Suppose that P is Poisson gr-simple. Then B is Poisson gr-simple if and only if B contains no homogeneous Poisson normal element $t^n + \sum_{i < n} b_i t^i$ with $n \geq 1$.*
- (3) *Suppose that P is Poisson gr-simple and $\text{char } k = 0$. Then B is Poisson gr-simple if and only if δ is not α -inner.*

Proof. For $a \in P$ and $n \geq 0$,

$$\{a, t^n\} = -nt^{n-1}(\alpha(a)t + \delta(a)). \quad (5)$$

(1) Let I be a graded Poisson ideal of P with $\alpha(I) + \delta(I) \subseteq I$. Then $IB = \bigoplus_n It^n$ is a graded ideal of B with $IB \cap P = I$. For $a \in P$ and $i \in I$, (5) gives $\{a, it^n\} = \{a, i\}t^n + i\{a, t^n\} \in IB$, and $\{t, it^n\} = (\alpha(i)t + \delta(i))t^n \in IB$. By the Leibniz rule IB is a Poisson ideal, so $I \in \{0, P\}$. If δ is α -inner, Lemma 4.2 gives a nonzero proper graded Poisson ideal.

(2) If such an element f exists, then Bf is a nonzero proper graded Poisson ideal. Conversely, let $J \neq 0$ be a graded Poisson ideal of B , let n be the least t -degree of a nonzero element of J , and let $L \subseteq P$ be the set of t^n -coefficients of the elements of J of t -degree at most n . As in Lemma 3.4, L is a graded ideal of P . If $g = ct^n + \dots \in J$ and $a \in P$, then by (5) the t^n -coefficient of $\{a, g\} \in J$ is $\{a, c\} - n\alpha(a)c$; since $n\alpha(a)c \in L$, also $\{a, c\} \in L$. So L is a nonzero graded Poisson ideal, and $L = P$. If $n = 0$, then $1 \in J$. Let $n \geq 1$. The component of degree γ^n of an element of J with t^n -coefficient 1 has the form $f = t^n + \sum_{i < n} b_i t^i$ and lies in J . For $a \in P$, the element $\{a, f\} + n\alpha(a)f \in J$ has t -degree less than n , so it vanishes. The

t^n -coefficient of $\{t, f\} = \sum_i (\alpha(b_i)t + \delta(b_i))t^i$ is $\alpha(b_{n-1})$, and its t^{n+1} -coefficient is $\alpha(1) = 0$; so $\{t, f\} - \alpha(b_{n-1})f \in J$ vanishes as well. By the Leibniz rule f is Poisson normal. Hence, if no such element exists, $J = B$.

(3) Necessity is part (1). Conversely, suppose that B is not Poisson gr-simple. By (2) there is a homogeneous Poisson normal $f = t^n + bt^{n-1} + \sum_{i < n-1} b_i t^i$ with $n \geq 1$. For $a \in P$ write $\{a, f\} = hf$. Comparing t -degrees gives $h \in P$, and the t^n -coefficients give $h = -n\alpha(a)$. By (5), the t^{n-1} -coefficient of $\{a, f\}$ is $-n\delta(a) + \{a, b\} - (n-1)b\alpha(a)$, while that of $-n\alpha(a)f$ is $-nb\alpha(a)$. Hence $n\delta(a) = \{a, b\} + b\alpha(a)$, that is, $\delta = \delta_u^P$ with $u = -n^{-1}b$. $\square$

For a commutative Noetherian $\mathbb{C}$ -algebra P with zero bracket and $\alpha = 0$, Jordan [16] shows that the prime spectrum of $P[t; \delta]$ is homeomorphic to the Poisson prime spectrum of $P[t; \delta]_P$.

4.1. Semiclassical limits. Let S be a ring and $h \in S$ a central non-zero-divisor such that S/hS is commutative, and consider the semiclassical limit S/hS (see Section 1). For Ore extensions in which $\sigma - \text{id}$ and δ take values in hR , the semiclassical limit is a Poisson polynomial ring [19, Proposition 4.1], [5].

Lemma 4.4. *Let $w \in S$, and let ϑ be an automorphism of S such that $sw = w\vartheta(s)$ and $\vartheta(s) - s \in hS$ for all $s \in S$. Then $\bar{w}$ is Poisson normal in S/hS .*

Proof. For $s \in S$, $sw - ws = w(\vartheta(s) - s)$, so $\{\bar{s}, \bar{w}\} = \bar{w} \overline{h^{-1}(\vartheta(s) - s)} \in \bar{w}(S/hS)$. $\square$

Let $S = R[z; \sigma, \delta]$, and let $h \in R$ be central in S and a non-zero-divisor, with R/hR commutative, $(\sigma - \text{id})(R) \subseteq hR$ and $\delta(R) \subseteq hR$. Put $P = R/hR$ and

$$\alpha_1(\bar{r}) = \overline{h^{-1}(\sigma(r) - r)}, \quad \delta_1(\bar{r}) = \overline{h^{-1}\delta(r)}.$$

Lemma 4.5. (1) α_1 and δ_1 are well defined, S/hS is the polynomial ring $P[\bar{z}]$, and $\{\bar{z}, \bar{r}\} = \alpha_1(\bar{r})\bar{z} + \delta_1(\bar{r})$. Hence $S/hS = P[z; \alpha_1, \delta_1]_P$.

(2) For $u \in R$, we have $\delta_u(R) \subseteq hR$ and $\overline{h^{-1}\delta_u(r)} = \delta_u^P(\bar{r})$, where δ_u^P is computed with α_1 . In particular, if $\delta = \delta_u$, then $\delta_1 = \delta_u^P$ is α_1 -inner.

Proof. (1) Since h is central, $zh = hz$, that is, $\sigma(h)z + \delta(h) = hz$; hence $\sigma(h) = h$ and $\delta(h) = 0$. For $r' \in R$ this gives $\sigma(hr') - hr' = h(\sigma(r') - r') \in h^2R$ and $\delta(hr') = h\delta(r') \in h^2R$, so both formulas are independent of the representative; division by h is unique because h is a non-zero-divisor. Since h is central and lies in R , $hS = \bigoplus_n hRz^n$, so $S/hS = \bigoplus_n P\bar{z}^n$. This ring is commutative, because $\bar{z}\bar{r} - \bar{r}\bar{z} = \overline{(\sigma(r) - r)z + \delta(r)} = 0$. Its semiclassical bracket extends that of P and satisfies

$$\{\bar{z}, \bar{r}\} = \overline{h^{-1}(zr - rz)} = \overline{h^{-1}(\sigma(r) - r)}\bar{z} + \overline{h^{-1}\delta(r)} = \alpha_1(\bar{r})\bar{z} + \delta_1(\bar{r}).$$

By [19, Theorem 2.1], α_1 is a Poisson derivation, δ_1 is a Poisson α_1 -derivation, and $S/hS = P[z; \alpha_1, \delta_1]_P$; compare [19, Proposition 4.1] and [5].

(2) We have $\delta_u(r) = (ur - ru) - (\sigma(r) - r)u$, and both terms lie in hR because R/hR is commutative. Hence

$$\overline{h^{-1}\delta_u(r)} = \{\bar{u}, \bar{r}\} - \alpha_1(\bar{r})\bar{u} = \delta_u^P(\bar{r}).$$

If $\delta = \delta_u$, then $\delta_1(\bar{r}) = \overline{h^{-1}\delta_u(r)} = \delta_u^P(\bar{r})$. $\square$

Example 4.6. Let $F = \mathbb{Q}[q^{\pm 1}]$, $h = q - 1$ and $R = F[x]$, graded by $\deg x = 1$ and $\deg q = 0$. Let $\sigma(x) = qx$ and $\delta(f) = (f(qx) - f(x))/x$, so that $\delta(x^m) = (q^m - 1)x^{m-1}$ for $m \geq 1$, and put $S = R[z; \sigma, \delta]$, graded by $\deg z = -1$. Thus

$$zx = qxz + (q - 1),$$

and h is central in S because $\sigma(q) = q$ and $\delta(q) = 0$. Specializing q to 2 gives the quantized Weyl algebra of Example 2.5. In S , the element $w = 1 + xz$ satisfies $wx = qxw$ and $zw = qwz$. Hence $sw = w\vartheta(s)$ for the automorphism ϑ of S with $\vartheta(x) = q^{-1}x$ and $\vartheta(z) = qz$, which preserves the defining relation. Since $\vartheta(x) - x$ and $\vartheta(z) - z$ lie in hS and $\vartheta(ab) - ab = (\vartheta(a) - a)\vartheta(b) + a(\vartheta(b) - b)$, we get $\vartheta(s) - s \in hS$ for every $s \in S$.

The semiclassical limit S/hS is $\mathbb{Q}[x, z]$ with $\{z, x\} = xz + 1$, that is, $B = \mathbb{Q}[x][z; \alpha, \delta_1]_p$ with $\alpha = x d/dx$ and $\delta_1 = d/dx$, and (α, δ_1) is (-1) -compatible. By Lemma 4.4, $\bar{w} = 1 + xz$ is Poisson normal, so B is not Poisson gr-simple. On the other hand, $\mathbb{Q}[x]$ with the zero bracket is (α, δ_1) -Poisson gr-simple, since no ideal (x^m) with $m \geq 1$ is δ_1 -stable, and δ_1 is not α -inner, because $\mathbb{Q}[x]_{-1} = 0$ and $\delta_1 \neq 0$ (Lemma 4.2(1)). This is the Poisson counterpart of Proposition 3.2. As in Remark 3.3, $1 + xz = x(z - u)$ with $u = -x^{-1}$, and δ_1 is α -inner over $\mathbb{Q}[x^{\pm 1}]$.

Remark 4.7. The conjugation phenomenon of Theorem A does not survive in a semiclassical limit. Let $S = R[z; \sigma, \delta]$ be graded as in Theorem 2.1 with $\deg z = \gamma$, let $h \in R_e$ be central in S and a non-zero-divisor, and suppose that $(\sigma - \text{id})(R) \subseteq hR$. If $\gamma\tau\gamma^{-1} \neq \tau$ and $r \in R_\tau$, then $\sigma(r) \in R_{\gamma\tau\gamma^{-1}}$, so $\sigma(r) - r \in hR$ has component $-r$ in degree τ . Since h is homogeneous, hR is graded, and $r \in hR$. Hence $R_\tau \subseteq hR$, and the grading of R/hR is supported on the centralizer of γ .

REFERENCES

- [1] Yassine Ait Mohamed. *Graded differential polynomial rings*. 2026. arXiv: [2606.23352](#).
- [2] S. A. Amitsur. “Derivations in simple rings”. In: *Proceedings of the London Mathematical Society* (3) 7 (1957), pp. 87–112. doi: [10.1112/plms/s3-7.1.87](#).
- [3] Kenneth S. Brown. *Cohomology of Groups*. Vol. 87. Graduate Texts in Mathematics. Springer-Verlag, 1982. doi: [10.1007/978-1-4684-9327-6](#).
- [4] Gérard Cauchon. “Idéaux bilatères et centre des anneaux de polynômes de Ore sur les anneaux quasi-simples”. In: *Séminaire d’Algèbre Paul Dubreil, 31ème année (Paris, 1977–1978)*. Vol. 740. Lecture Notes in Mathematics. Berlin: Springer, 1979, pp. 397–407.
- [5] Eun-Hee Cho and Sei-Qwon Oh. “Semiclassical limits of Ore extensions and a Poisson generalized Weyl algebra”. In: *Letters in Mathematical Physics* 106.7 (2016), pp. 997–1009. doi: [10.1007/s11005-016-0856-4](#). arXiv: [1602.02195](#).
- [6] Chen-Lian Chuang and Yuan-Tsung Tsai. “On the structure of semi-invariant polynomials in Ore extensions”. In: *Journal of Algebra* 322.7 (2009), pp. 2464–2491. doi: [10.1016/j.jalgebra.2009.06.022](#).
- [7] Wagner Cortes and Miguel Ferrero. “Principal ideals in Ore extensions”. In: *Mathematical Journal of Okayama University* 46 (2004), pp. 77–84. doi: [10.18926/mjou/33921](#).
- [8] Julián Martín Eisenschlos. “3-Calabi–Yau algebras from Steiner systems”. Tesis de Licenciatura. Universidad de Buenos Aires, 2013.
- [9] K. R. Goodearl. “Prime ideals in skew polynomial rings and quantized Weyl algebras”. In: *Journal of Algebra* 150.2 (1992), pp. 324–377. doi: [10.1016/S0021-8693\(05\)80036-5](#).
- [10] K. R. Goodearl and R. B. Warfield Jr. *An Introduction to Noncommutative Noetherian Rings*. 2nd ed. Vol. 61. London Mathematical Society Student Texts. Cambridge University Press, 2004. doi: [10.1017/CB09780511841699](#).
- [11] Can Hatipoğlu and Christian Lomp. *Iterated Hopf Ore extensions over group rings*. 2026. arXiv: [2602.10850](#).
- [12] Roozbeh Hazrat. *Graded Rings and Graded Grothendieck Groups*. Vol. 435. London Mathematical Society Lecture Note Series. Cambridge: Cambridge University Press, 2016.

- [13] D. A. Jordan. “Ore extensions and Jacobson rings”. PhD thesis. University of Leeds, 1975.
- [14] D. A. Jordan. “A simple localization of the quantized Weyl algebra”. In: *Journal of Algebra* 174.1 (1995), pp. 267–281. doi: [10.1006/jabr.1995.1128](https://doi.org/10.1006/jabr.1995.1128).
- [15] D. A. Jordan. “Normal elements of degree one in Ore extensions”. In: *Communications in Algebra* 30.2 (2002), pp. 803–807. doi: [10.1081/AGB-120013183](https://doi.org/10.1081/AGB-120013183).
- [16] David A. Jordan. “Ore extensions and Poisson algebras”. In: *Glasgow Mathematical Journal* 56.2 (2014), pp. 355–368. doi: [10.1017/S0017089513000293](https://doi.org/10.1017/S0017089513000293).
- [17] T. Y. Lam and A. Leroy. “Homomorphisms between Ore extensions”. In: *Azumaya Algebras, Actions, and Modules*. Ed. by D. Haile and J. Osterburg. Vol. 124. Contemporary Mathematics. American Mathematical Society, 1992, pp. 83–110.
- [18] T. Y. Lam, K. H. Leung, A. Leroy, and J. Matczuk. “Invariant and semi-invariant polynomials in skew polynomial rings”. In: *Ring Theory 1989 (Ramat Gan and Jerusalem, 1988/1989)*. Ed. by L. Rowen. Vol. 1. Israel Mathematical Conference Proceedings. 1989.
- [19] Stéphane Launois and César Lecoutre. “A quadratic Poisson Gel’fand–Kirillov problem in prime characteristic”. In: *Transactions of the American Mathematical Society* 368 (2016), pp. 755–785.
- [20] André Leroy and Jerzy Matczuk. “The extended centroid and X-inner automorphisms of Ore extensions”. In: *Journal of Algebra* 145.1 (1992), pp. 143–177.
- [21] J. C. McConnell and J. C. Robson. *Noncommutative Noetherian Rings*. Vol. 30. Graduate Studies in Mathematics. With the cooperation of L. W. Small. American Mathematical Society, 2001.
- [22] Constantin Năstăsescu and Freddy Van Oystaeyen. *Methods of Graded Rings*. Vol. 1836. Lecture Notes in Mathematics. Springer, 2004. doi: [10.1007/b94904](https://doi.org/10.1007/b94904).
- [23] Sei-Qwon Oh. “Poisson polynomial rings”. In: *Communications in Algebra* 34.4 (2006), pp. 1265–1277.
- [24] Johan Öinert. “Simple group graded rings and maximal commutativity”. In: *Operator Structures and Dynamical Systems*. Vol. 503. Contemporary Mathematics. Providence, RI: American Mathematical Society, 2009, pp. 159–175. doi: [10.1090/conm/503/09899](https://doi.org/10.1090/conm/503/09899).
- [25] Johan Öinert, Johan Richter, and Sergei D. Silvestrov. “Maximal commutative subrings and simplicity of Ore extensions”. In: *Journal of Algebra and Its Applications* 12.4 (2013), p. 1250192. doi: [10.1142/S0219498812501927](https://doi.org/10.1142/S0219498812501927).
- [26] Øystein Ore. “Theory of non-commutative polynomials”. In: *Annals of Mathematics* 34.3 (1933), pp. 480–508. doi: [10.2307/1968173](https://doi.org/10.2307/1968173).
- [27] A. N. Panov. “Ore extensions of Hopf algebras”. In: *Mathematical Notes* 74.3 (2003), pp. 401–410. doi: [10.1023/A:1026115004357](https://doi.org/10.1023/A:1026115004357).
- [28] Y. Shen and Y. Guo. “Nakayama automorphisms of graded Ore extensions of Koszul Artin–Schelter regular algebras”. In: *Journal of Algebra* 579 (2021), pp. 114–151. doi: [10.1016/j.jalgebra.2021.02.033](https://doi.org/10.1016/j.jalgebra.2021.02.033).

DÉPARTEMENT DE MATHÉMATIQUES, UNIVERSITÉ DE SHERBROOKE, 2500 BD DE L’UNIVERSITÉ, SHERBROOKE, QC, J1K 2R1, CANADA

Email address: yassine.ait.mohamed@usherbrooke.ca